

Q1. Smart University Payroll Management System

Score: 25.00 TC: 3/3 passed

CODE

```
import java.util.*;

class Main
{
    public static String readInput(Scanner sc)
    {
        String s = "";

        while(sc.hasNextLine())
        {
            s = sc.nextLine().trim();

            if(!s.isEmpty())
            {
                return s;
            }
        }

        return s;
    }

    public static void main(String args[])
    {
        Scanner sc = new Scanner(System.in);

        int n = Integer.parseInt(readInput(sc));

        Employee emp[] = new Employee[n];

        for(int i=0;i<n;i++)
        {
            String type = readInput(sc);

            int id = Integer.parseInt(readInput(sc));
            String name = readInput(sc);
            String department = readInput(sc);
            double basicSalary = Double.parseDouble(readInput(sc));

            if
            {
                int teachingHours = Integer.parseInt(readInput(sc));
                double incentivePerHour = Double.parseDouble(readInput(sc));

                emp[i] = new TeachingFaculty(id,name,department,
                    basicSalary,teachingHours,incentivePerHour);
            }
            else if(type.equalsIgnoreCase("TechnicalStaff"))
            {
                int overtimeHours = Integer.parseInt(readInput(sc));
                double overtimeRate = Double.parseDouble(readInput(sc));

                emp[i] = new TechnicalStaff(id,name,department,
                    basicSalary,overtimeHours,overtimeRate);
            }
            else if(type.equalsIgnoreCase("VisitingFaculty"))
            {
                int lecturesHandled = Integer.parseInt(readInput(sc));
                double paymentPerLecture = Double.parseDouble(readInput(sc));

                emp[i] = new VisitingFaculty(id,name,department,
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                    basicSalary,lecturesHandled,paymentPerLecture);
            }
        }

        String search = readInput(sc);

        if(search.equalsIgnoreCase("Search Employee ID:"))
        {
            search = readInput(sc);
        }

        int searchId = Integer.parseInt(search);

        Employee highest = emp[0];
        double totalSalary = 0;

        System.out.println("Employee Details");
        System.out.println():
```

Q2. Smart Vehicle Insurance Claim System

Score: 25.00 TC: 3/3 passed

CODE

```
import java.util.*;

class Main
{
    public static String readInput(Scanner sc)
    {
        String s="";

        while(sc.hasNextLine())
        {
            s=sc.nextLine().trim();

            if(!s.isEmpty())
                return s;
        }

        return s;
    }

    public static void main(String args[])
    {
        Scanner sc=new Scanner(System.in);

        int n=Integer.parseInt(readInput(sc));

        Vehicle vehicle[]=new Vehicle[n];

        for(int i=0;i<n;i++)
        {
            String type=readInput(sc);

            String vehicleNumber=readInput(sc);
            String ownerName=readInput(sc);
            double insuredAmount=Double.parseDouble(readInput(sc));

            if(type.equalsIgnoreCase("Car"))
            {
                vehicle[i]=new Car(vehicleNumber,ownerName,insuredAmount);
            }
            else if(type.equalsIgnoreCase("Bike"))
            {
                vehicle[i]=new Bike(vehicleNumber,ownerName,insuredAmount);
            }
            else if(type.equalsIgnoreCase("Truck"))
            {
                vehicle[i]=new Truck(vehicleNumber,ownerName,insuredAmount);
            }
        }
    }
}
```

```
        if(vehicle[i].calculateClaimAmount()>highest.calculateClaimAmount())
        {
            highest=vehicle[i];
        }
    }

    for(int i=0;i<n;i++)
    {
        System.out.println("Vehicle : " + vehicle[i].getVehicleNumber());
        System.out.println("Claim Amount : " + vehicle[i].calculateClaimAmount());
        System.out.println();
    }

    System.out.println("Highest Claim Vehicle");
    System.out.println(highest.getVehicleNumber());
    System.out.println(highest.calculateClaimAmount());
    System.out.println();

    boolean found=false;

    for(int i=0;i<n;i++)
    {
        if(vehicle[i].getVehicleNumber().equalsIgnoreCase(search))
        {
            System.out.println("Vehicle Found");
            System.out.println("Owner : " + vehicle[i].getOwnerName());
            System.out.println("Claim Amount : " + vehicle[i].calculateClaimAmount());

            found=true;
            break;
        }
    }

    if(!found)
    {
        System.out.println("Vehicle Not Found");
    }
}

class Vehicle
{
    private String vehicleNumber;
    private String ownerName;
    private double insuredAmount;
```